Supplementary Materials: Search Strategy

Detailed Search Strategies by Database

The following search strategies were developed to map the interdisciplinary evidence linking deep space radiation exposure with oncogenesis, genomic susceptibility, and circulatory risk. Searches were conducted from inception through ~~April 15, 2026~~September 21, 2026.

1. Ovid MEDLINE

Platform: Ovid

Date Searched: ~~April 15, 2026~~September 21, 2026

Strategy:

1. exp Space Flight/ OR exp Extraterrestrial Environment/ OR astronaut*.mp.
2. exp Cosmic Radiation/ OR exp Linear Energy Transfer/ OR (high-LET OR HZE OR galactic cosmic ray* OR solar particle event*).mp.
3. 1 AND 2
4. exp Neoplasms, Radiation-Induced/ OR exp Cardiovascular Diseases/ OR (cancer risk OR NSCR OR oncogenesis).mp.
5. exp Genomic Instability/ OR exp DNA Damage/ OR Clonal Hematopoiesis.mp. OR (non-targeted effect* OR bystander effect* OR CHIP).mp.
6. Heavy Ion Radiotherapy/ OR (hadrontherapy OR carbon ion radiotherapy).mp.
7. 4 OR 5 OR 6
8. 3 AND 7
9. limit 8 to english language

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2. Embase

Platform: Elsevier

Date Searched: ~~April 15, 2026~~September 21, 2026

Strategy:

#1 'space flight'/exp OR 'astronaut'/exp OR 'extraterrestrial environment'/exp
#2 'cosmic radiation'/exp OR 'heavy ion radiation'/exp OR 'linear energy transfer'/exp OR 'high-LET':ti,ab,kw OR 'galactic cosmic ray*':ti,ab,kw
#3 #1 AND #2
#4 'radiation carcinogenesis'/exp OR 'cardiovascular disease'/exp OR 'cancer risk':ti,ab,kw
#5 'genomic instability'/exp OR 'DNA damage'/exp OR 'clonal hematopoiesis'/exp OR 'non-targeted effect*':ti,ab,kw

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#6 'carbon ion radiation'/exp OR 'hadrontherapy':ti,ab,kw
#7 #4 OR #5 OR #6
#8 #3 AND #7
#9 #8 AND [english]/lim

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3. Cochrane Central Register of Controlled Trials (CENTRAL)
Platform: Wiley
Date Searched: ~~April 15, 2026~~September 21, 2026
Strategy:

#1 [mh "Space Flight"] OR [mh "Cosmic Radiation"]
#2 ("space radiation" OR "galactic cosmic rays" OR "high-LET" OR "HZE particles"):ti,ab,kw
#3 #1 OR #2
#4 ("cancer risk" OR "clonal hematopoiesis" OR "DNA damage" OR "hadrontherapy" OR "NSCR"):ti,ab,kw
#5 #3 AND #4

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4. Scopus
Platform: Elsevier
Date Searched: ~~April 15, 2026~~September 21, 2026
Strategy:

TITLE-ABS-KEY (("space flight" OR astronaut) AND ("space radiation" OR "galactic cosmic ray*" OR "high-LET" OR "solar particle event*")) AND TITLE-ABS-KEY ("cancer risk" OR "NSCR" OR "clonal hematopoiesis" OR "genomic susceptibility" OR "DNA damage" OR "non-targeted effect*" OR "cardiovascular" OR "hadrontherapy") AND LIMIT-TO (LANGUAGE , "English")*

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5. Web of Science (Core Collection)
Platform: Clarivate
Date Searched: ~~April 15, 2026~~September 21, 2026
Strategy:

TS=(("space flight" OR astronaut) AND ("space radiation" OR "galactic cosmic ray*" OR "high-LET" OR "solar particle event*" OR "HZE particles")) AND TS=("cancer risk" OR "NSCR" OR "clonal hematopoiesis" OR "genomic susceptibility" OR "DNA damage" OR "non-targeted effect*" OR "cardiovascular" OR "hadrontherapy")*
Refined by: LANGUAGES: (ENGLISH)
